

EXPERIMENT 18

Create a GUI calculator where buttons perform addition and subtraction.

Code:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

class Calculator extends JFrame implements ActionListener {

    JTextField tf1, tf2, tfResult;
    JButton btnAdd, btnSub;

    Calculator() {
        setTitle("Simple Calculator");
        setSize(400, 250);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLayout(new GridLayout(4, 2, 10, 10));

        // Input fields
        add(new JLabel("Enter Number 1:"));
        tf1 = new JTextField();
```

```
add(tf1);
```

```
add(new JLabel("Enter Number 2:"));
```

```
tf2 = new JTextField();
```

```
add(tf2);
```

```
// Buttons
```

```
btnAdd = new JButton("Addition");
```

```
btnSub = new JButton("Subtraction");
```

```
add(btnAdd);
```

```
add(btnSub);
```

```
add(new JLabel("Result:"));
```

```
tfResult = new JTextField();
```

```
tfResult.setEditable(false);
```

```
add(tfResult);
```

```
btnAdd.addActionListener(this);
```

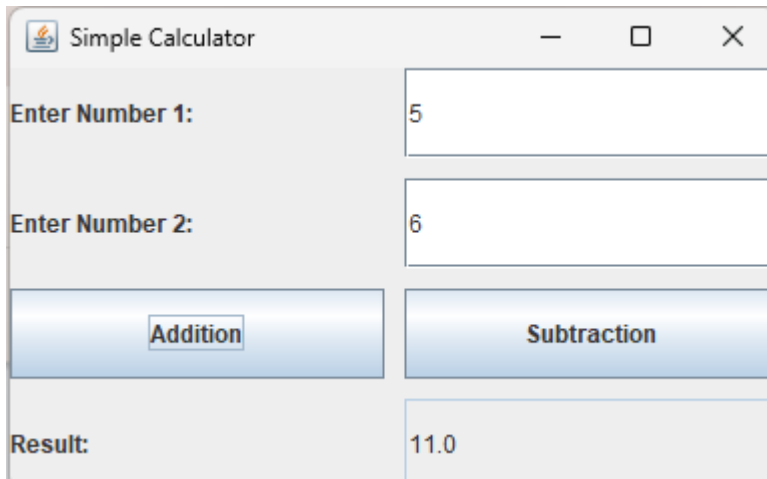
```
btnSub.addActionListener(this);
```

```
setVisible(true);
```

```
}
```

```
public void actionPerformed(ActionEvent e) {  
    double n1 = Double.parseDouble(tf1.getText());  
    double n2 = Double.parseDouble(tf2.getText());  
  
    if (e.getSource() == btnAdd) {  
        tfResult.setText(String.valueOf(n1 + n2));  
    }  
  
    if (e.getSource() == btnSub) {  
        tfResult.setText(String.valueOf(n1 - n2));  
    }  
}  
  
public static void main(String[] args) {  
    new Calculator();  
}  
}
```

Output:



Create a Bank Balance Calculator GUI application using Java Swing where the user enters the initial balance and transaction amount. Use buttons to perform addition (deposit) and subtraction (withdrawal) and display the updated balance

Code:

```
import javax.swing.*;
```

```
import java.awt.*;
```

```
import java.awt.event.*;
```

```
class BankBalance extends JFrame implements ActionListener {
```

```
    JTextField tfBalance, tfAmount, tfResult;
```

```
    JButton btnDeposit, btnWithdraw;
```

```
    BankBalance() {
```

```
setTitle("Bank Balance Calculator");  
  
setSize(450, 250);  
  
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);  
  
setLayout(new GridLayout(4, 2, 10, 10));
```

```
// Initial balance
```

```
add(new JLabel("Initial Balance:"));  
  
tfBalance = new JTextField();  
  
add(tfBalance);
```

```
// Transaction amount
```

```
add(new JLabel("Transaction Amount:"));  
  
tfAmount = new JTextField();  
  
add(tfAmount);
```

```
// Buttons
```

```
btnDeposit = new JButton("Deposit");  
  
btnWithdraw = new JButton("Withdraw");
```

```
add(btnDeposit);  
  
add(btnWithdraw);
```

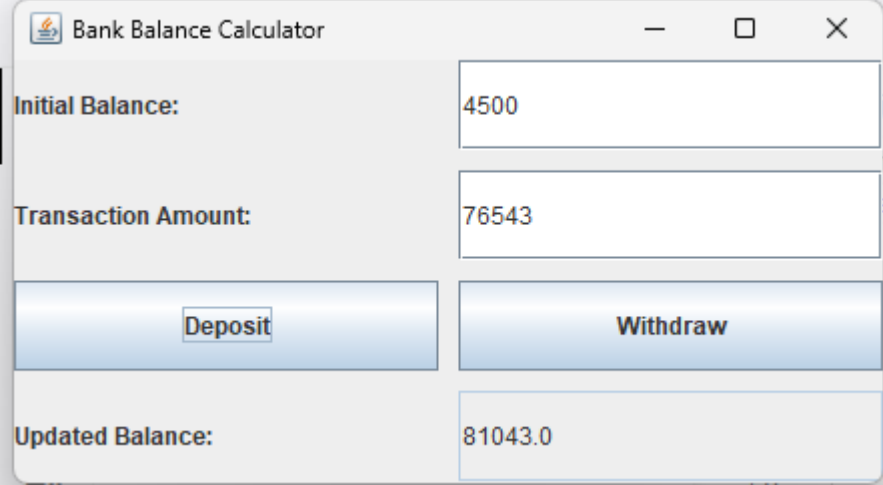
```
// Updated balance
```

```
add(new JLabel("Updated Balance:"));
```

```
tfResult = new JTextField();  
tfResult.setEditable(false);  
add(tfResult);  
  
btnDeposit.addActionListener(this);  
btnWithdraw.addActionListener(this);  
  
setVisible(true);  
}  
  
public void actionPerformed(ActionEvent e) {  
  
    double balance = Double.parseDouble(tfBalance.getText());  
    double amount = Double.parseDouble(tfAmount.getText());  
  
    // Deposit adds money  
    if (e.getSource() == btnDeposit) {  
        balance = balance + amount;  
    }  
  
    // Withdrawal subtracts money  
    if (e.getSource() == btnWithdraw) {  
        balance = balance - amount;  
    }  
}
```

```
        tfResult.setText(String.valueOf(balance));  
    }  
  
    public static void main(String[] args) {  
        new BankBalance();  
    }  
}
```

Output:



The image shows a Java Swing window titled "Bank Balance Calculator". It contains three text input fields and two buttons. The first field, labeled "Initial Balance:", contains the value "4500". The second field, labeled "Transaction Amount:", contains the value "76543". Below these fields are two buttons: "Deposit" and "Withdraw". The "Deposit" button is highlighted with a blue gradient. Below the buttons is a third text input field labeled "Updated Balance:" which contains the value "81043.0".

Initial Balance:	4500
Transaction Amount:	76543
DepositWithdraw	
Updated Balance:	81043.0